Decentralised Single-Index Bandits

Learning an unknown reward function without a server: theorems and findings

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Problem statement

Setting

N agents on a graph $\mathcal{G}$, **no server**. Each round agent i sees K contexts $x \in \mathbb{R}^d$, pulls one, observes $y = f(\langle x, \theta^* \rangle) + \eta$. Unknown, shared θ^* and f (f possibly non-monotone). Goal: small network regret $R_T(N)$, little communication.

Difficulties

- **Two unknowns:** the direction needs pooled data; the shape of f needs per-bin counts.
- **No coordinator:** totals are built over the graph, $\approx$ diameter rounds late.
- **Stale or averaged counts** break the UCB confidence argument.
- **Communication:** sharing tables every round costs $\Theta(|E|N_b T)$ scalars.

What prior work misses

Single-index bandits are single-agent; decentralised bandits assume finite arms, linear or kernel models. No decentralised single-index bandit, and no graph-dependent lower bound for this class.

Literature survey

Work	What it does	Use / gap
Dey, Bhore & Ghosh (2026)	one agent, non-monotone f; optimal $\tilde{\Theta}(T^{2/3})$	our base algorithm; no network
Kang et al. (ICLR 2026)	ESTOR (monotone f), GSTOR ($T^{3/4}$)	baselines; one agent
Wang et al. (ICLR 2020)	bandits *with a server*; event-triggered sync	our trigger; needs a server
Landgren (2016); DDUCB (NeurIPS 2019)	decentralised K-armed; consensus, gossip	mixing baselines; finite arms
DLUCB (AAAI 2021)	decentralised *linear* bandits; delayed, consistent statistics	our epoch idea; linear model
Coop-KernelUCB (ICML 2020)	kernels, message passing; graph lower bound *open*	relaying; costly in d

Gap: an unknown index *and* an unknown non-monotone link, learned by agents that talk only to neighbours.

Contributions ⋆

1. **New problem:** the first decentralised single-index bandit. Agents share θ^* and f and talk only over a graph.
2. **Lower bounds:** no serverless algorithm beats $\Omega((NT)^{2/3})$ (Thm 5). On rings and paths it must also pay $\Omega(N \min(T, D)^{1/3})$ (Thm 7, light cones).
3. **Upper bound for the algorithm we run:** the published-only relay has regret $\tilde{O}\big((dNT)^{2/3} + KF'\big)$ (Thm 27): the server's rate plus an *additive, horizon-free* graph term set by the farness F'. It holds for every trigger threshold γ, so γ trades regret against communication.
4. **Communication and robustness:** logarithmic in T for a fixed grid (Thm 29), $O(N_b + N)$ memory per agent. Directed, delayed and time-varying graphs change only the delivery time, and our bound $B(N-1)$ is tight.
5. **Experiments:** 37 studies with paired trials and 95% CIs, run in the order the problems came up. The measured graph cost has the proved shape, bin owners remove what is left of it, and we compare with prior papers' algorithms on our problem.

Recap: single-index bandits

Model

Each round an agent sees K arms $x_{t,a} \in \mathbb{R}^d$, pulls one, and observes

$$\boxed{y_t = f(\langle x_t, \theta_* \rangle) + \eta_t}$$

θ_* and f unknown, f possibly non-monotone.

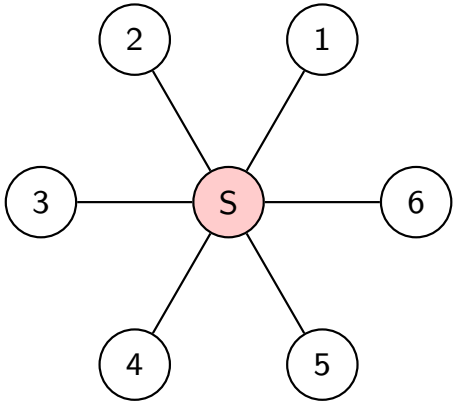
ZoomSIB-UCB (Dey, Bhore & Ghosh 2026): optimal $\tilde{\Theta}(T^{2/3})$

- **Phase 1:** explore; estimate θ_* with the Stein estimator $\hat{v} = \frac{1}{T_0} \sum_t y_t S(x_t) \propto \theta_*$, a *sample mean*.
- **Phase 2:** project $z = x^\top \hat{\theta}$, cut the line into bins, run *sleeping* UCB over the bins available each round. State: (n_j, S_j) per bin.

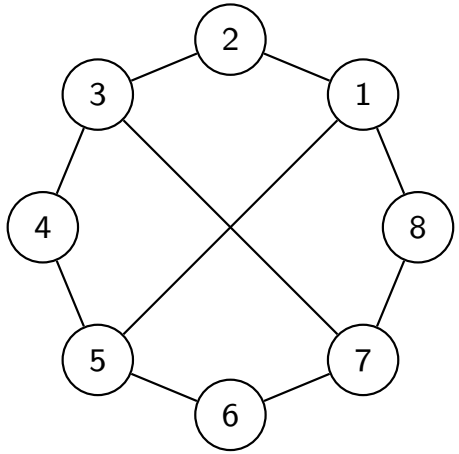
Our question

N agents share θ_* and f but **no server**: they talk only to graph neighbours.

With a server vs without a server



Server (star)
one coordinator sees all statistics



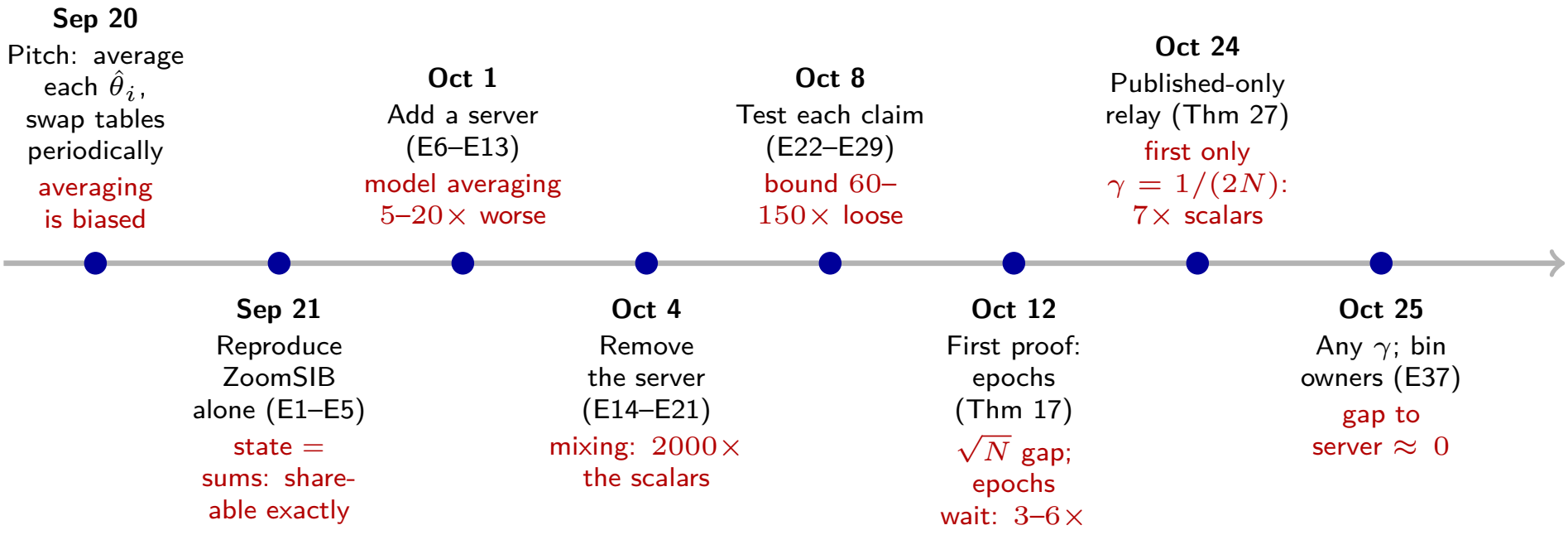
Serverless (graph G)
diameter D, spectral gap $1 - \lambda$

Why serverless?

No trusted coordinator, or no backbone: sensor networks, robot/drone swarms, vehicular networks, peer-to-peer recommendation, consortia that do not share raw data.

Network regret $R_T(N) = \sum_{i,t} \left[f(\langle x^*_{i,t}, \theta_* \rangle) - f(\langle x_{i,t}, \theta_* \rangle) \right]$

Our path in one picture



Each step starts from what failed in the step before. Red: what went wrong, or what it cost.

Experimental setup

Instances (as in Dey, Bhore & Ghosh)

$d = 10$, $K = 20$ contexts $\sim \mathcal{N}(0, I)$, noise $\sigma = 0.1$. Links: quadratic $-(z-1)^2 + 1$ (non-monotone), zigzag $\sin z + 0.3z$, logistic (monotone).

Networks

$N = 4$–32 agents (16 by default) on complete, expander, hypercube, torus, ring, path graphs ($D = 1$–15), plus failing, delayed, time-varying and directed links.

Protocol

- $T \leq 20{,}000$ per agent ($NT \leq 320$k); pooled Phase 1 $= 960\,(NT/160\text{k})^{2/3}$; $\Delta = (NT)^{-1/3}$; $\gamma = 0.5$ unless stated.
- 10–20 **paired** trials (the same seeds for every method), mean $\pm$ 95% CI.
- Metrics: network regret, *excess* over the server, scalars sent.
- References: server, independent agents. Baselines: consensus / gossip / push-sum protocols; prior algorithms with ideal relaying.

Step 1: reproduce ZoomSIB with one agent (E1–E5)

What we checked

- Stein direction error falls like $n^{-1/2}$: slope -0.50 for $d = 5$–40 (E1).
- ZoomSIB beats LinUCB, ESTOR and GSTOR on all three non-monotone links. ESTOR is even worse than LinUCB on the quadratic link: the price of assuming f is monotone (E2).
- Log-log regret slope 0.29–0.60, below the theory's $2/3$ (E3).
- Phase 1 needs only $\approx 2\%$ of the horizon; adaptive stopping exits at 1.7% (E5).

What we took from it

The learning state is a direction estimate and a per-bin table (n_j, S_j), and both are sums. Agents can add sums exactly, so we decided to share sums. Our pitch had planned to average each agent's fitted estimate instead.

Step 2: add a server, and our first plan fails (E6–E13)

We tried	What happened	So we
Average each agent's normalised $\hat{\theta}_i$ (the pitch's plan)	direction error 0.332 vs 0.196 for pooled Stein sums, under covariate shift (E10)	pool the *sums*, then normalise once
FL model averaging of the bin tables (FedAvg, distillation)	table 5–$20\times$ *worse than not syncing*: agents weighted by total pulls, not pulls in the bin (E11)	send exact (j, n_j, S_j)
Sync the tables every round	1.57M scalars	sync only when a bin's news is big
Event trigger $\gamma = 0.5$	same regret ($5{,}187$ vs $5{,}155$) with 30k scalars (E8)	keep $\gamma = 0.5$ as the trigger

With a server, collaboration pays

Network regret $\propto N^{0.48}$ when both phases are shared, $N^{0.77}$ with Phase 1 only, $N^{1.04}$ for independent agents (E6).

Step 3: remove the server: mix or relay?

Phase 1: build $\sum_k b_k$ over the graph

- DGD, decentralised FedAvg: stay **biased** by 4–46%; on the star, regret $103{,}555$ vs the server's $9{,}674$.
- Gradient tracking: unbiased, 208k scalars.
- **Spanning tree:** exact, 2.6k scalars.

Phase 2: share the bin tables

- Consensus, push-sum: server's regret, but 140–285M scalars.
- Burst gossip: $+4$–13%; effective-sample-size counts no better than naive ones.
- One-hop neighbours only: $+20$–91%.
- **Event-triggered flooding:** $+2$–5%, 70k scalars.

What we learned

Averaging (mixing) only converges towards the sum. Relaying exact, sparse records delivers the sum itself, a few rounds late, and gives the same regret with about $2000\times$ fewer scalars. Every later design relays.

Step 3, continued: a misleading win

Surprise: the network beat the server

With adaptive Phase-1 stopping, push-sum on a torus had regret $5{,}761$, *below* the server's $9{,}719$ (E16). Agents that disagree keep exploring longer, so the gain came from the stopping rule and not from the network.

Fix: compare at a fixed T_0 (every later experiment)

Then relays track the diameter ($+1\%$ at $D{=}1$, $+15\%$ at $D{=}15$) and mixing tracks the spectral gap.

Second surprise: acceleration is fragile (E18)

Chebyshev gossip: $\times 2$ at 30% link failures, $\times 7$ at 90%. Flooding: $+27\%$ at 90%, because it only needs every message to arrive eventually.

Design after Step 3

Spanning-tree Phase 1 + event-triggered flood relay: $\times 1.030$ of the server with 60k scalars, best trade-off of 28 designs (E20).

Step 4: test each claim before proving it (E22–E28)

Claim	Measured	What it told us
A server = one agent with NT pulls	within 0–10%; independent agents 2.3–5.1× worse (E22)	the target rate is $(NT)^{2/3}$
Losing the server is cheap	$+1.5\%$ (D=1) to $+9.1\%$ (D=15); grows like $T^{0.13}$ (E23)	our sketched bound $N \cdot D \cdot N_b$ was 59–149× too big: the graph cost should not grow with T
Longer exploration helps on graphs	no: the best T_0 does not move; agents freezing *different* $\hat{\theta}_i$ cost ×1.25–1.34; one agreement pass fixes it (E28)	agreement matters more than exploration length (Thm 33)
Compress to save traffic	Phase 1 never (top-25%: ×5.96); γ cliff $0.5 \to 1$: $\times 1.004 \to \times 1.52$ (E26, E27)	keep Phase 1 exact; $\gamma \leq 0.5$

What was left to prove

(1) What no algorithm can beat. (2) A delay term that does not grow with T. (3) That we lose nothing in the confidence widths.

Theory: setting of the proofs

Assumptions (DBG's, with one made explicit)

- Contexts i.i.d., coordinates κ-sub-Gaussian, $\mathbb{E}[S_j(X)^2] \le M$; $|f| \le L_f$, $|f'| \le L_{f'}$ on $|z| \le X_{\max}$; $\mu^* = \mathbb{E} f'(\langle X, \theta^* \rangle) > 0$.
- Noise i.i.d. and σ-sub-Gaussian. DBG's Stein lemma already needs i.i.d. samples, and their hard instance has $\mathcal{N}(0,1)$ noise.

Communication: messages move one hop per round

A flooded message from agent l reaches agent i after $d(i,l)$ rounds, and everyone within the **delivery time** Λ ($= D$, the diameter, on a static graph). The regret bound uses the distances through the **farness** $F' = \max_i \sum_l \max(d(i,l), 1) \le ND$; robustness results replace them by Λ.

Order of the theory (`docs/proofs/serverless.pdf`)

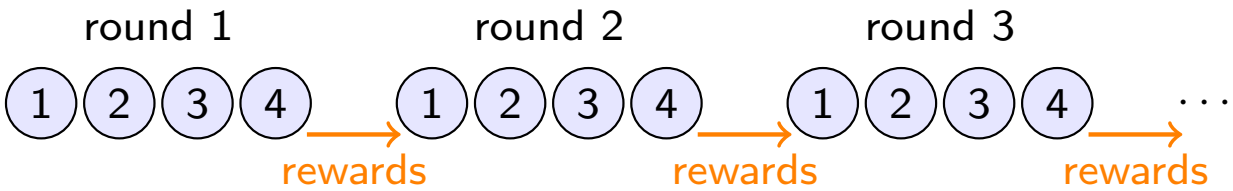
No algorithm beats (Thm 5, 7) $\rightarrow$ ours achieves (Thm 27) $\rightarrow$ it sends (Thm 29) $\rightarrow$ bad networks (Lemma 31) $\rightarrow$ agents must agree (Thm 33).

Theorem 5: no serverless algorithm beats $(NT)^{2/3}$

Theorem

For any N-agent algorithm, on any graph, even one where every agent sees all data after every round: $\sup_{\text{instances}} \mathbb{E}\, R_T(N) \geq c\,(NT)^{2/3}$.

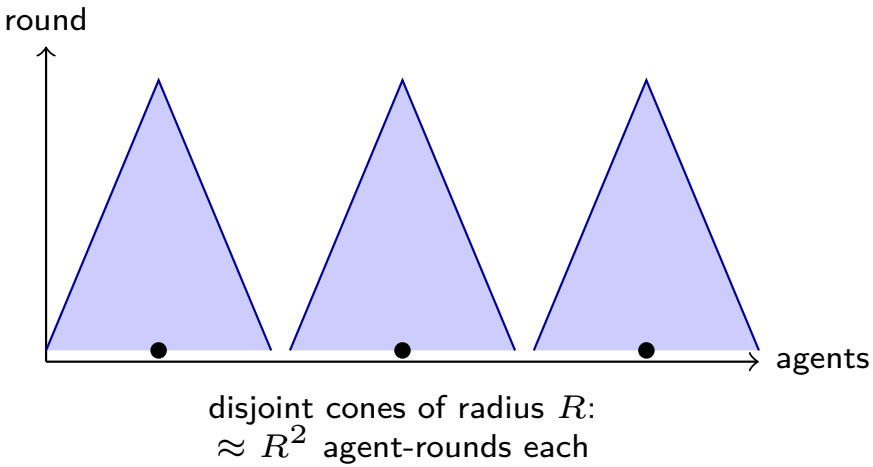
Proof idea: one agent can simulate the whole network.



- Single-agent round $u = (t-1)N + i$ plays what agent i would play in round t.
- It **withholds** round t's rewards until all N agents of round t have acted, then runs one communication step.
- Same law of the history $\Rightarrow$ network regret $=$ regret of one agent over NT rounds $\geq c(NT)^{2/3}$ (DBG Thm 5.1).

Hard instances: DBG's at horizon NT ($d = 1$, Gaussian contexts, $K \approx (NT)^{1/3} \log NT$ arms).

Theorem 7: the graph does cost regret (light cones)



Lemma 6 (light cones are closed)

Information moves one hop per round, so the actions inside a cone depend only on data inside it.

Theorem 7 and Corollary 8

Each cone acts like one agent with $n \approx R^2$ pulls, so it pays $\Omega(n^{2/3})$. One common hard instance makes all the cones pay at once. On a ring or a path:

$$\sup \mathbb{E}\, R_T(N) \;\geq\; c \max\left\{ (NT)^{2/3}, \quad N \min(T, D)^{1/3} \right\}.$$

- The second term wins when $T \lesssim \sqrt{ND}$: there, **no network on a ring can match one agent with NT pulls**.
- Longer horizons: upper graph term KF' (Thm 27) vs this $ND^{1/3}$. Linear in distance necessary? Open: a cone already learns at rate $n^{2/3}$ on its own data.

Step 5: the first proof breaks; epochs fix it

The gap in the first proof

In the flood relay, an agent uses its own unpublished rewards. A reward changes the agent's next pulls, which change *when* its trigger fires, i.e. when the pull is published. Each agent's view became a union of N per-agent prefixes, and the proof lost a factor $\sqrt{N}$ in the confidence width, which breaks the matching rate.

First fix: synchronised epochs (Theorems 17, 22; backup)

Every agent acts on the *same* delayed table plus its own pulls: one common prefix plus one own block, so the server's width. Proved: $\tilde{O}\big((dNT)^{2/3} + N\Lambda K\big)$, horizon-free. The measured excess divided by NDK is 0.86–1.15, so the bound has the right shape.

But it is slow (E31)

$+5\%$ to $+59\%$ over the server, against $+1.5\%$ to $+9\%$ for the relay we ran: 3–$6\times$ the extra regret. Every agent waits for the farthest agent ($\approx 4D$ rounds).

Step 6: the algorithm we now run

Every agent i runs the same code

Phase 1: learn the direction

1: **for** $t \le T_0$: pull a random arm; $b_i \leftarrow b_i + \varphi_\tau\big(y\, S(x)\big)$ ▷ truncated Stein sum

2: flood b_i; wait Λ rounds ▷ every agent now holds all N sums

3: $\hat{\theta} \leftarrow \sum_k b_k / \|\sum_k b_k\|_1$ ▷ identical everywhere; equals the server's

Phase 2: learn f with sleeping UCB on bins of width Δ along $\langle x, \hat{\theta}\rangle$

4: **for** each round t:

5: $(n_j, S_j) \leftarrow \sum_k$ freshest *published* table of k ▷ own table too, once published

6: pull an arm in the available bin with the largest $S_j/n_j + c_\delta/\sqrt{n_j}$

7: **if** own unpublished pulls in some bin j exceed $\gamma \max(n_j, 1)$: flood own table

Why this design

- Phase 1 is **exact**: Stein sums are sufficient statistics, so flooding them gives the server's $\hat{\theta}$.
- Phase 2 relays **exact, sparse** tables, published only when a bin grows by a factor γ.
- Acting only on *published* data is what makes Theorem 27 provable.

Step 6: Theorem 27, regret of the algorithm we run

Theorem 27 (published-only relay, any $\gamma > 0$, w.p. $\geq 1 - 4\delta$)

$$R_T(N) \leq \underbrace{2L_f N(T_0+\Lambda)}_{\text{Phase 1}} + \underbrace{4\sqrt{2}\, c'_\delta \sqrt{(1+N\gamma)N_b NT}}_{\text{UCB}} + \underbrace{2L_{f'}\Delta NT}_{\text{bin width}} + \underbrace{4L_{f'} X_{\max}\varepsilon_0 NT}_{\text{direction error}}$$
$$+ \underbrace{2L_f\left(4F' \min(K, N_b) + N_b(6\ell' + N\gamma + 1)\right)}_{\text{delay: the only graph term}}$$

Rates ($\Delta = (NT)^{-1/3}$, T_0 balanced)

- $\gamma = \gamma_0/N$: $R_T(N) = \tilde{O}\left((dNT)^{2/3} + KF'\right)$; matches Thm 5 when $KF' \leq (NT)^{2/3}$.
- constant γ (we run 0.5): $\tilde{O}\left((dNT)^{2/3} + \sqrt{1+N\gamma}\,(NT)^{2/3} + KF'\right)$, $N\times$ fewer publications.

Reading the bound

The graph adds a term that does not grow with T and is set by the sum of distances: star $F' \approx 2N$, ring $F' \approx ND/2$, path $F' \leq ND$.

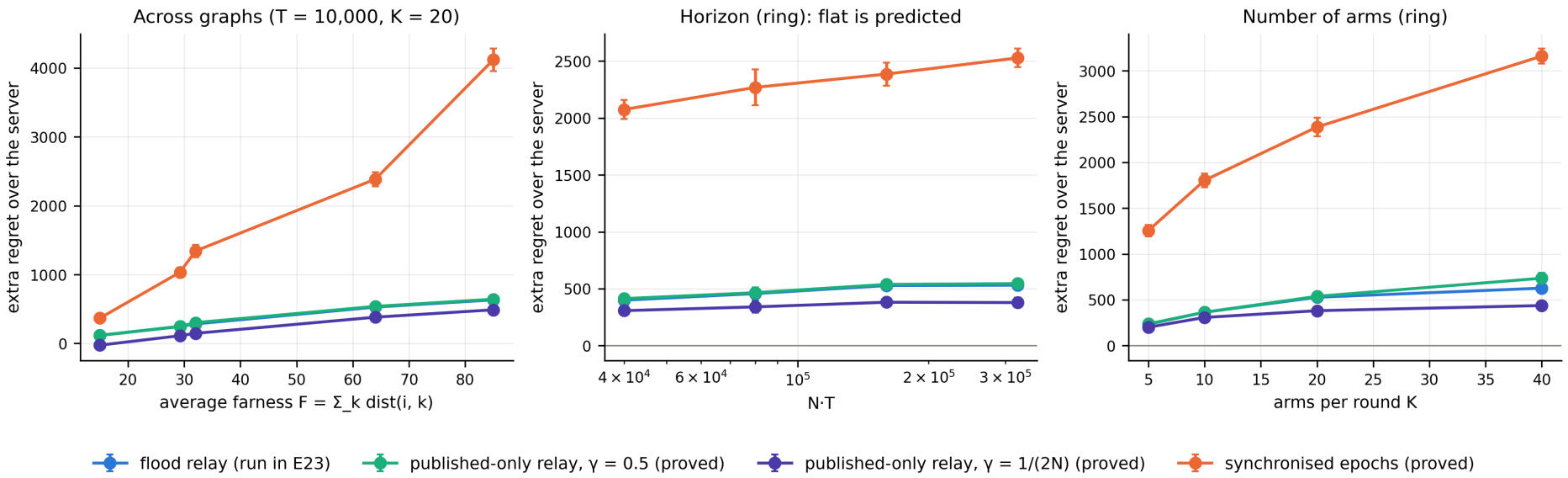
Why Theorem 27 holds: three ideas

1. **One prefix, no $\sqrt{N}$ (Lemmas 24, 25).** Agents act only on *published* records, their own included. So a reward cannot change *when* its pull is published, and anything it influences reaches every agent later (triangle inequality). Each agent's view is one prefix of a fixed order: a martingale, so the confidence width is the server's.
2. **Views nest (Lemma 26).** What agent l saw by round s, agent i sees by round $s + d(i, l)$. So l's unpublished pulls are at most $\gamma\times$ a count that i *also* sees, and after a short lag i sees a $1/(2(1+N\gamma))$ fraction of the true count, whatever γ is.
3. **Only K arms (Lemma 21).** A pull into bin j needs an arm in bin j. In the $\leq F'$ agent-rounds that i cannot yet see, only K arms per agent-round exist, so in total $\approx KF'$ pulls are unseen, not F' per bin, and the term does not grow with T.

Step 6, continued: from $\gamma = 1/(2N)$ to any γ

Our first version of this proof needed $N\gamma \leq 1/2$, i.e. $\gamma = 1/(2N)$: 7–8$\times$ the scalars of the relay we ran (E36). Idea 2 removed the restriction, so $\gamma = 0.5$, the setting we run, is now proved.

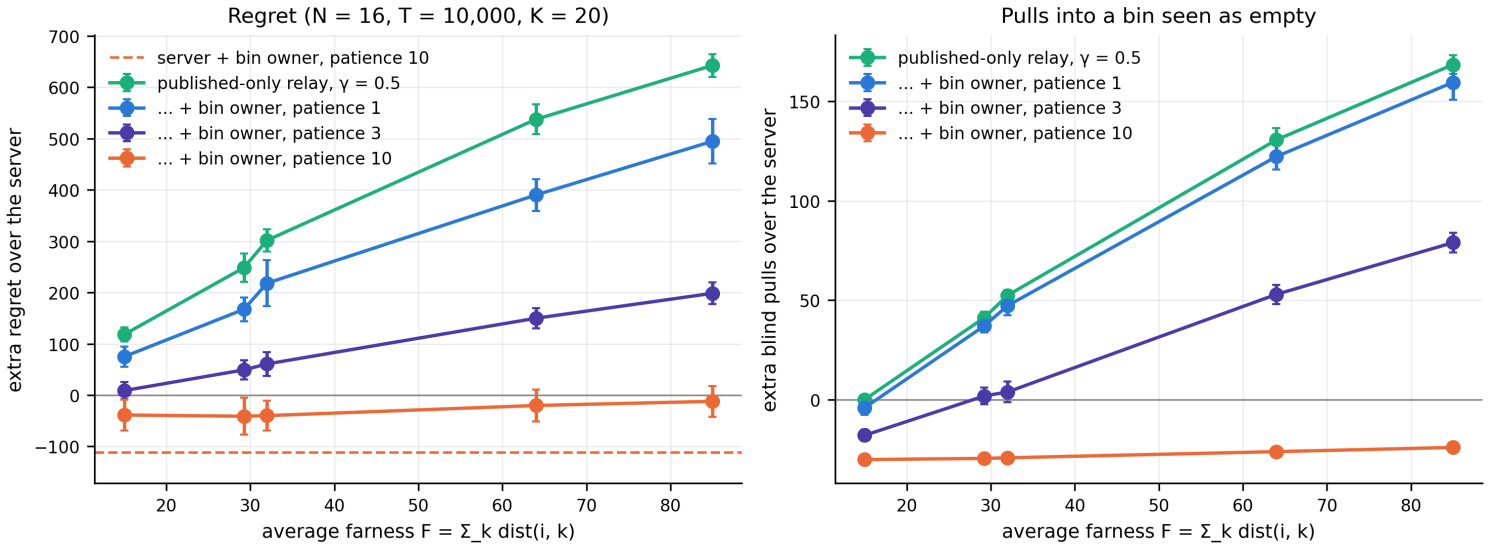
Evidence: the graph's cost has the proved shape



- **Across graphs:** the relays follow the farness F (excess/$KF = 0.38$–0.47 for our relay at $\gamma = 0.5$); the epoch protocol follows NDK (ratio 0.86–1.15).
- **Horizon:** $8\times$ longer T raises the excess only $\times 1.3$, as Theorem 27 says (horizon-free up to logs).
- **Arms:** the excess grows like $K^{0.4}$–$K^{0.55}$, slower than the bound's linear K, so that factor is loose (open).
- γ **dial:** $\gamma = 1/(2N)$ lowers the excess (ring 381 vs 538) for $7.6\times$ more scalars.

$N = 16$, quadratic link, 10 paired trials. The server reference uses $\gamma = 0.5$, so $\gamma = 1/(2N)$ can sit slightly below it.

Step 7: bin owners remove the rest of the gap (E37)



- **Where the KF' term comes from:** far-apart agents explore the same *empty* bin before they hear of each other. Such blind pulls grow with the farness (server 412, ring 543, path 581).
- **Bin owners:** each bin has an owner fixed by a shared hash, so *no messages*. An agent leaves an empty bin to its owner until it has seen that bin available A times (patience).
- **Patience 10:** excess over the server -41 to -12 on every graph (was 118–643); blind pulls back to the server's level; $1.6\times$ the scalars (285k), still $4.8\times$ below $\gamma = 1/(2N)$.
- **Caveat:** the rule also helps a server (-112, dashed). Against that server the relay pays 71–100, nearly flat in F'. Not yet proved (open): decisions use counts only, so Lemmas 24–25 hold, but held pulls need their own term.

Evidence: collaboration pays without a server (E19)

Network regret for $N = 4 \ldots 32$ agents (quadratic link, $T = 10{,}000$, fixed pooled Phase-1 size), fitted as $\propto N^{\alpha}$:

	$N = 4$	8	16	32	α
server	3,707	4,152	5,839	8,888	0.43
our relay, complete	3,724	4,175	5,926	9,067	0.44
our relay, hypercube ($D = \log_2 N$)	3,726	4,189	6,183	9,466	0.46
our relay, ring ($D = N/2$)	3,727	4,219	6,514	11,615	0.55
Chebyshev gossip, ring	3,825	4,427	9,483	22,389	0.87
independent agents	8,194	17,224	33,310	63,971	0.98

- Our relay keeps the server's $N^{0.43}$ scaling on well-connected graphs; at $N = 32$ the hypercube is within 7% of the server, the ring within 31%.
- Independent agents scale almost linearly; mixing on the ring loses most of the benefit.

Published-only relay, $\gamma = 0.5$; 6 paired trials.

Theorem 29: communication

Theorem (published-only relay; the same count holds for epochs)

Each flooded message crosses each edge at most once, and each agent publishes at most $N_b\big(1+\log_{1+\gamma}(NT)\big)$ times, so the number of scalars is at most

$$\underbrace{2|E|N(d{+}1)}_{\text{Phase 1}} + \underbrace{2|E|\,N\,N_b\big(1+\log_{1+\gamma}(NT)\big)(3N_b{+}1)}_{\text{Phase 2}}.$$

- **Why logarithmic:** between two publications triggered by a bin, its count grows by a factor $> 1+\gamma$.
- For a fixed grid this is **logarithmic in** T, against $\Theta(|E|N_bT)$ for per-round sharing.

Choosing γ (Thm 27)

$\gamma = \gamma_0/N$ gives the optimal rate but $\approx N\times$ more publications ($\tilde{O}(|E|N^2(NT)^{2/3})$ at the rate-optimal grid). A constant γ publishes $N\times$ less and the bound pays $\sqrt{1+N\gamma}$. Measured (E36, ring): $\gamma = 0.5$ sends 180k scalars for $+2\%$ regret; $\gamma = 1/32$ sends 1.37M.

Evidence: relaying vs mixing

Design (torus, $N = 16$, same T_0)	**Regret ÷ server**	**Scalars sent**
Server + event-triggered sync	1.000	57k
Spanning tree + published-only relay, $\gamma = 0.5$	1.034	64k
Flooding / consensus / push-sum every round	1.02	9.5M–295M

- **Same regret, about** $2{,}200$–$4{,}600\times$ **fewer scalars** than consensus/push-sum (140–295M scalars).
- On regret *alone*, mixing is not worse: consensus every round is as good or slightly better on well-connected graphs. Its cost is communication.
- Relaying depends on the distances, mixing on the spectral gap, which degrades when links fail (E18).

E20: ring, torus, expander × i.i.d./covariate-shift contexts, 6 paired trials; the table is the geometric mean over the 6 cells.

Robustness: only the delivery time changes

Network	Delivery time Λ
static, connected	D
directed, strongly connected	directed D
link delay $\leq \tau$	τD
union of every B rounds connected	$B(N-1)$, **tight**
subgraph stable for B rounds	D_B (its diameter)
links fail w.p. q	$\approx (D + \log NT)/(1-q)$

Why (Lemma 31, Cor 32)

A slow or failing link delays records but never corrupts them. The $(NT)^{2/3}$ term never changes; only the additive graph term grows (distances $\to \Lambda$, so $F' \to N\Lambda$).

Measured (E18, expander, $N = 16$): links fail w.p. q, regret $\div$ server

	$q = 0.3$	$q = 0.6$	$q = 0.9$
Published-only relay	1.07	1.10	1.26
Running consensus	1.03	1.11	1.50
Chebyshev (accelerated) gossip	2.10	3.54	7.31

Theorem 33: agents must agree on the direction

Theorem

If agent k bins with its own $\hat{\theta}_k$ and $\max_k \|\hat{\theta}_k - \hat{\theta}\|_1 \le \eta_{\text{agree}}$ ($\hat{\theta}$ = exact pooled), the regret grows by at most

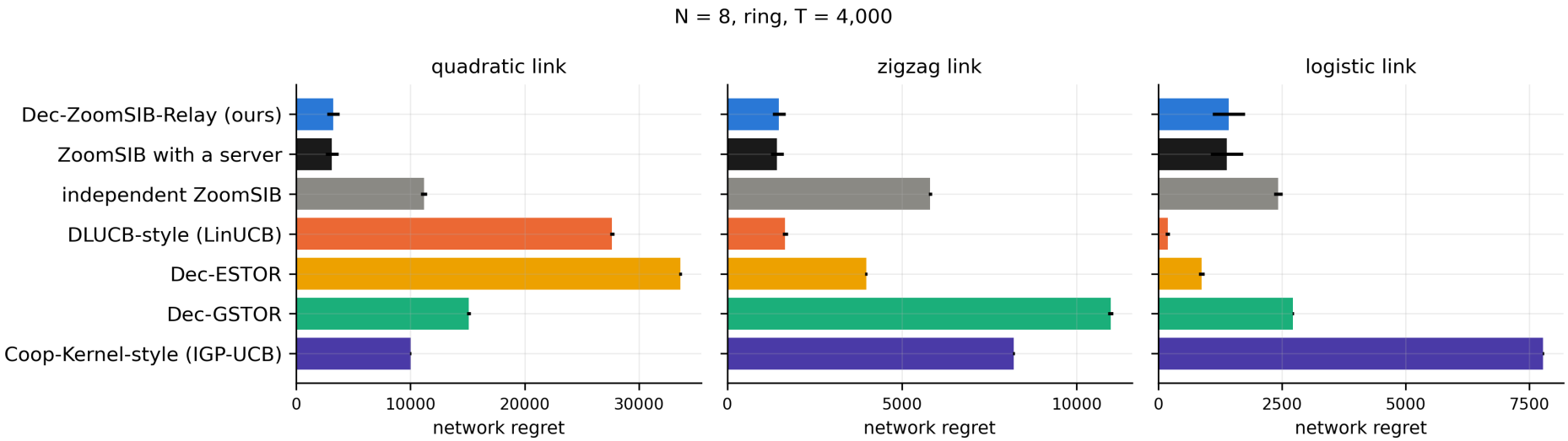
$$4L_{f'} X_{\max} \eta_{\text{agree}} NT.$$

- **Why:** directions enter the proof only through the projection error. Pooling a bin across agents still pools pulls whose means lie in the same interval.
- **Linear in NT:** even small disagreement hurts. It matters once $X_{\max}\eta_{\text{agree}}NT \sim (NT)^{2/3}$.
- **Fix:** one extra flood of the local Stein sums gives every agent the exact direction ($\eta_{\text{agree}} = 0$), for Λ rounds and $2|E|N(d{+}1)$ scalars.

Measured

Consensus Phase 1: $\times 1.25$ (ring), $\times 1.34$ (path). One flooding pass: $\times 1.08$, $\times 1.09$.

Results: prior papers' algorithms on our problem



- **Non-monotone link (quadratic):** ours is best, and the gap grows with T. Kernel UCB: $1.24\times$ ours at $T{=}1000$, $3.1\times$ at $T{=}4000$. Linear / monotone methods: 8.5–$10\times$.
- **Communication:** 16–$70\times$ fewer scalars than relaying raw data.
- **Monotone link (logistic):** LinUCB $(0.13\times)$ and ESTOR $(0.61\times)$ win, because they assume the monotonicity that holds here and we do not.
- **Mostly increasing (zigzag $= \sin z + 0.3z$):** LinUCB $0.54\times$ at $T{=}1000$, $1.12\times$ at $T{=}4000$.

$N = 8$, ring, $d = 10$, $K = 20$, 10 paired trials. Baselines get *ideal* relaying (every observation, exact, at graph-distance delay), at least as much information as their own gossip.

What the failures taught us

Failure	Cost	Lesson
averaging estimates or models	$\times 1.7$ direction error; 5–$20\times$ table error	share exact sufficient statistics
periodic or per-round sync	$\approx 50\times$ the traffic	trigger on news: γ
consensus / push-sum	$\approx 2000\times$ the scalars	relay, do not mix
accelerated gossip	$\times 7$ at 90% link failures	need only eventual delivery
adaptive stopping on graphs	fake win over the server	compare at a fixed T_0; agree first
bound $N \cdot D \cdot N_b$	59–$149\times$ loose	charge only first visits: horizon-free
first relay proof	$\sqrt{N}$ wider confidence	act only on published data
synchronised epochs	3–$6\times$ extra regret	do not wait for the farthest agent
proof only at $\gamma = 1/(2N)$	$7\times$ the scalars	views nest $\Rightarrow$ any γ
duplicate first visits	the whole KF' gap	bin owners, no messages

Conclusion and future work

Conclusion

- Without a server we get the server's rate plus a graph cost that does not grow with T and is set by the farness.
- We prove the protocol we run (agents act only on published data), for any trigger γ.
- The measured cost has the proved shape. Relaying matches consensus with orders of magnitude fewer scalars, and loses only 26% when 90% of link uses fail.

Limitations

- Graph lower bound $N \min(T, D)^{1/3}$ vs upper KF': a gap. Bounds linear in K; we measure $\approx \sqrt{K}$.
- Synthetic data only; private Phase 2 is open. The newest proofs are still being checked.

Future work

Prove the bin-owner rule (E37); close the graph gap; gap-dependent K; private Phase 2; heterogeneous and asynchronous agents.

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Thank You!